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$$\lim_{x\to\infty}\left(\frac{ax^2+bx+c}{ax^2+dx+f}\right)^x = \lim_{x\to\infty}\left(\frac{x^2}{x^2}\right)^x = \left(\frac{\infty}{\infty}\right)^\infty = 1^\infty$$
$$\Rightarrow \lim_{x\to\infty}\left(\frac{ax^2+bx+c}{ax^2+dx+f}\right)^x = \lim_{x\to\infty}\left(\frac{ax^2+dx+f}{ax^2+dx+f}+\frac{bx+c-dx-f}{ax^2+dx+f}\right)^x$$
$$= \lim_{x\to\infty}\left(1+\frac{(b-d)x+(c-f)}{ax^2+dx+f}\right)^x = e^{\lim\limits_{x\to\infty} x\frac{(b-d)x+(c-f)}{ax^2+dx+f}} = e^{\lim\limits_{x\to\infty}\frac{(b-d)x^2+(c-f)x}{ax^2+dx+f}}$$
$$= e^{\lim\limits_{x\to\infty}\frac{(b-d)x^2}{ax^2}} = e^{\frac{b-d}{a}}$$

References